

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times int \times int \Rightarrow State \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{noncrit2} : Loc \\ \text{wait2} : Loc \\ \text{crit2} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit1}, p^2, y^1, y^2) \rightarrow \text{state}(\text{wait1}, p^2, y^{11}, y^2) \quad [y^{11} = (y^2 + 1)] \\ 2 : & \text{state}(\text{wait1}, p^2, y^1, y^2) \rightarrow \text{state}(\text{crit1}, p^2, y^1, y^2) \quad [y^2 = 0 \vee (y^1 < y^2)] \\ 3 : & \text{state}(\text{crit1}, p^2, y^1, y^2) \rightarrow \text{state}(\text{noncrit1}, p^2, y^{11}, y^2) \quad [y^{11} = 0] \\ 4 : & \text{state}(p^1, \text{noncrit2}, y^1, y^2) \rightarrow \text{state}(p^1, \text{wait2}, y^1, y^{21}) \quad [y^{21} = (y^1 + 1)] \\ 5 : & \text{state}(p^1, \text{wait2}, y^1, y^2) \rightarrow \text{state}(p^1, \text{crit2}, y^1, y^2) \quad [y^1 = 0 \vee (y^2 < y^1)] \\ 6 : & \text{state}(p^1, \text{crit2}, y^1, y^2) \rightarrow \text{state}(p^1, \text{noncrit2}, y^1, y^{21}) \quad [y^{21} = 0] \\ 7 : & \text{state}(p^1, p^2, y^1, y^2) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit1}, \text{crit2}, y^1, y^2) \rightarrow \text{error} \end{array} \right.$$

$\{9: \text{state}(\text{noncrit1}, \text{noncrit2}, 0, 0) \Rightarrow \text{success}\}$

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bitvec_4 \times bitvec_4 \Rightarrow State \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{noncrit2} : Loc \\ \text{wait2} : Loc \\ \text{crit2} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll}
1 : & \text{state}(\text{noncrit1}, p2, y1, y2) \rightarrow \text{state}(\text{wait1}, p2, y11, y2) \quad [y11 = \text{ite}(y2 = \#x0, (y2 \text{bvadd} \#x1), (y2 \text{bvadd} \#x2))] \\
2 : & \text{state}(\text{wait1}, p2, y1, y2) \rightarrow \text{state}(\text{crit1}, p2, y1, y2) \quad [y2 = \#x0 \vee y2 = y1 \text{bvadd} \#x2] \\
3 : & \text{state}(\text{crit1}, p2, y1, y2) \rightarrow \text{state}(\text{noncrit1}, p2, y11, y2) \quad [y11 = \#x0] \\
4 : & \text{state}(p1, \text{noncrit2}, y1, y2) \rightarrow \text{state}(p1, \text{wait2}, y1, y21) \quad [y21 = \text{ite}(y1 = \#x0, (y1 \text{bvadd} \#x1), (y1 \text{bvadd} \#x2))] \\
5 : & \text{state}(p1, \text{wait2}, y1, y2) \rightarrow \text{state}(p1, \text{crit2}, y1, y2) \quad [y1 = \#x0 \vee y1 = y2 \text{bvadd} \#x2] \\
6 : & \text{state}(p1, \text{crit2}, y1, y2) \rightarrow \text{state}(p1, \text{noncrit2}, y1, y21) \quad [y21 = \#x0] \\
7 : & \text{state}(p1, p2, y1, y2) \rightarrow \text{success} \\
8 : & \text{state}(\text{crit1}, \text{crit2}, y1, y2) \rightarrow \text{error}
\end{array} \right.$$

{ 9: state(noncrit1, noncrit2, #x0, #x0) => success }